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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2025

EC2102-1 – ELECTROMAGNETIC WAVE THEORY
(For EC,AC,VL)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- The boundary condition at the interface of two media states that the tangential components of the electric field
 - Must be continuous
 - Must be discontinuous
 - Must be zero
 - Are independent of media properties
- In a dielectric material, the electric flux density/displacement vector (D) is defined as
 - $D = \epsilon_0 E$
 - $D = \epsilon E$
 - $D = \epsilon_0 E + P$
 - $D = E/\epsilon$
- If the magnetic flux density B is changing with time in a region of free space, the induced electric field E is
 - Conservative
 - Constant
 - Zero
 - Non-conservative
- Which of the following is a scalar quantity?
 - Electric field intensity
 - Electric potential
 - Electric flux density
 - Magnetic flux
- The divergence theorem relates _____.
 - Line integral and surface integral
 - Line integral and volume integral
 - Surface integral and volume integral
 - Volume integral and curl
- In a perfect conductor, the electric field inside is
 - Equal to the surface charge density
 - Infinity
 - Zero
 - Equal to the external field
- According to Faraday's Law, an increasing magnetic flux through a closed loop will induce
 - A constant current
 - An electric field opposing the change in flux
 - A magnetic field in the opposite direction
 - No effect
- Maxwell's first equation is derived from _____.
 - Gauss's Law for electric fields
 - Faraday's Law
 - Ampere's Circuital Law
 - Gauss's Law for magnetism
- The velocity of a wave travelling in the air medium without transmission lines or waveguides(wireless) is _____.
 - 6×10^8 m/s
 - 3×10^8 m/s
 - 1.5×10^8 m/s
 - 9×10^8 m/s
- Skin depth phenomenon is found in _____ materials.
 - Insulators
 - Dielectrics
 - Conductors
 - Semiconductors

11. Which of the following is the correct relation between wavelength and the phase constant of a wave?
 A) Phase constant = $2\pi/\text{wavelength}$ B) Phase constant = $2\pi \times \text{wavelength}$
 C) Phase constant = $1/(2\pi \times \text{wavelength})$ D) Phase constant = $\text{wavelength}/2\pi$
12. Which of Maxwell's equations states that magnetic monopoles do not exist?
 A) Gauss's Law for electricity B) Gauss's Law for magnetism
 C) Ampere's Law D) Faraday's Law
13. Maxwell modified Ampere's law by introducing _____.
 A) The concept of magnetic monopole B) The relationship between voltage and resistance
 C) The concept of wave propagation D) The displacement current term
14. The propagation constant (γ) in a conducting medium is given by
 A) $\gamma = \alpha + j\beta$ B) $\gamma = j\omega\epsilon$
 C) $\gamma = j\beta$ D) $\gamma = j\alpha$
15. In a perfect conductor, the skin depth is _____.
 A) Dependent on permittivity B) Zero
 C) Equal to wavelength D) Infinite
16. If the electric field vector remains in a fixed plane while propagating, the wave is said to be
 A) Circularly polarized B) Randomly polarized
 C) Linearly polarized D) Elliptically polarized
17. The Poynting vector represents
 A) The direction and magnitude of power flow in an electromagnetic wave B) The rate of change of electric field
 C) The current density in a conductor D) The impedance of free space
18. On a lossless transmission line, the power delivered to the load is maximum when
 A) $Z_L \neq Z_0$ B) $Z_L = \infty$
 C) $Z_L = Z_0$ D) $Z_L = 0$
19. The center of the Smith Chart represents _____.
 A) Open circuit B) Infinite impedance
 C) Short circuit D) Matched impedance condition
20. If the normalized impedance is inside the unit circle on the Smith Chart, it represents _____.
 A) A capacitive load B) An inductive load
 C) A matched impedance D) A resistive load only

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) State and explain Gauss law and hence derive the Maxwell's first equation for electrostatics.	8	L*2	1	1
	b) Magnetic field intensity in free space is given by $H = 10 \rho^2 a_\phi$ A/m. Determine current density J and total current over the circular surface for radial distance = 1m, for all ϕ and $z=0$.	8	L3	2	1
2.	a) Determine magnetic field intensity at a point due to a current in a straight conductor of infinite length. Use Biot Savart's law	8	L2	2	1
	b) Three-point charges $2\mu\text{C}$, $4\mu\text{C}$, and $8\mu\text{C}$ are located at (0, 0, 0), (0, 0, 1), and (1, 0, 0), respectively. Find the energy in the system.	8	L3	1	1
3.	a) Derive the expressions for boundary conditions between a conductor and free space.	8	L2	1	1
	b) Evaluate closed line integral of magnetic field intensity about rectangular path P1(2,3,4) to P2(4,3,4) to P3(4,3,1) to P4(2,3,1) to P1 and $H=8z a_x - 4x^3 a_z$ A/m.	8	L3	2	1

Unit – II

4. a) Obtain the expression for displacement current density starting from Ampere's circuital law. 8 L2 3 1
- b) Define skin depth. A steel pipe is constructed of a material for which $\mu_r = 180$ and $\sigma = 4 \times 10^6$ S/m. If the total current carried by pipe is $2\cos 104\pi t$ Amperes, find skin depth. 8 L3 4 1
5. a) State and explain Faraday's law of electromagnetic induction and obtain the differential form of Faraday's law. 8 L2 3 1
- b) A plane EM wave travelling in lossless dielectric medium in positive z-direction with $\mu_r = 1$ and $\epsilon_r = 3$. The EM wave has peak electric field intensity of 16 V/m. Determine
- Velocity of the wave
 - Characteristic impedance
 - Peak value of magnetic field
 - Peak value of pointing vector
 - Phase constant if $f = 6580$ MHz
 - Wavelength
- 8 L3 4 1
6. a) Derive the expressions for a total power and Poynting vector of the electromagnetic wave. 8 L2 3 1
- b) Determine whether the given pair of fields satisfy Maxwell's equations in the region where conductivity is 0, $\epsilon = 2.5 \epsilon_0$ and $\mu = 10 \mu_0$. Given $E = 3y a_y$ and $H = 4x a_x$. 8 L3 4 1

Unit – III

7. a) Derive the voltage equation of a transmission line. 8 L2 5 1
- b) A load of $Z_R = 115 - j75 \Omega$ terminates a lossless 100Ω line. Use Smith chart to determine standing wave ratio, input impedance of 0.2λ long line and distance from load to the first voltage maximum. 8 L3 5 1
8. a) What is the relation between VSWR and reflection coefficient? Draw the standing wave pattern for open circuit, closed circuit, and matched transmission line conditions. 8 L2 5 1
- b) Derive the expression for input impedance of a transmission line. 8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
